Research Concept Note

Title: Towards Semantic Bibliometrics: A Distributed Heterogeneous Graph Architecture for Intent-Aware Impact Evaluation and GraphRAG

1. Abstract & Motivation

Traditional bibliometric indices (e.g., the h-index) and advanced multi-hop graph metrics (e.g., hp-frac) quantify scientific impact by evaluating the structural topology of citation networks. While metrics like hp-frac successfully mitigate flaws such as hyperauthorship by leveraging fractional allocation across citation paths, they currently possess a "semantic blindspot." Existing topological models treat all citations uniformly, failing to mathematically differentiate between a citation that fundamentally extends a methodology versus one that refutes a flawed dataset.

At the same time, processing deep NLP across massive citation networks is computationally expensive. To resolve this, we propose a scalable system that uses a Heterogeneous Graph Neural Network (HGNN) to classify the intent behind each citation. This allows us to compute the **Semantic hp-frac**: a novel metric that evaluates true scientific contribution rather than relying on easily inflated raw citation counts. Finally, this intent-aware graph will power an interactive dashboard, guiding AI search tools along verified citation paths to generate highly accurate and context-aware literature reviews.

2. Primary Research Objectives

1. **Formulate the Semantic hp-frac Index:** Evolve the existing fractional multi-hop metric by substituting binary citation edges with continuous, semantic weights derived from citation intent classification.
2. **Architect a Feature-Rich Heterogeneous Semantic Graph:** Move beyond sparse metadata representations by integrating "Section-Aware" NLP. We will initialize ‘Paper’ nodes using deep, rich text embeddings (via SciBERT) of the explicitly extracted *Methodology* and *Abstract* sections, rather than just paper titles. This rich initialization, combined with temporal decay dynamics, allows the Heterogeneous Graph Neural Network (HGNN) to calculate attention weights based on deep methodological overlap rather than superficial keyword matching.
3. **Engineer a Distributed Big Data Pipeline:** Decouple the deep learning inference from graph traversal by materializing semantic weights into optimized Parquet columnar storage, allowing Apache Spark to execute complex metric calculations and GraphRAG queries with ultra-low latency.

3. Proposed Methodology

3.1. Data Ingestion & Section-Aware NLP

To capture deep contextual features, standard PDF parsing is insufficient. We will utilize GROBID to parse unstructured academic PDFs into TEI-encoded XML. This allows for the extraction of **Section-Aware NLP features**. For every citation edge, the system extracts the raw citation context as well as the spatial metadata (e.g., identifying if the citation occurred in the "Methodology" vs. "Related Work" section). Node features will be initialized using domain-specific SciBERT embeddings of the extracted abstracts.

3.2. Citation Intent Classification (Training Strategy)

To train the edge-weighting mechanism, we propose a two-phase methodology:

- **Phase 1: Domain-Specific Bootstrapping (PoC):** To rapidly prototype the graph within our specific target domain (e.g., Indic NLP or neuro-symbolic AI), we will utilize Large Language Models (LLMs) via weak supervision. We will programmatically prompt an LLM to label a localized dataset of 5,000 citation sentences into a strict taxonomy (Extends, Baseline, Background, Refutes).
- **Phase 2: Formal Evaluation on Benchmark Datasets (Gold Standard):** For the formal research publication, relying solely on LLM-generated labels is methodologically insufficient. We will benchmark and fine-tune our architecture against established, manually annotated academic datasets. Specifically, we will utilize the **SciCite** dataset, which contains 11K manually annotated citations across 3 intents (background information, method, and result comparison) , and the **ACL-ARC** dataset, comprising 2K citations annotated for 6 intents (background, uses, compare/contrast, motivation, extends, future work). This dual-phase approach ensures both domain flexibility and academic rigor.

3.3. HGNN Architecture & Temporal Decay Dynamics

The primary computational engine will be developed utilizing the PyTorch Geometric framework to facilitate high-performance graph representation learning.

- **Node Feature Engineering:** We define heterogeneous entities comprising **Paper** nodes, featurized via high-dimensional text embeddings, and **Author** nodes, featurized through the aggregation of historical publication metadata.
- **Edge Taxonomy:** The architecture models relational dependencies through multi-typed **Cites** and **Writes** edges, representing the core interactions within the scholarly ecosystem.
- **Modeling Temporal Decay:** To capture the evolution of scientific consensus, **Cites** edges will incorporate a dynamic temporal delta $\Delta t = (Year_{Citing} - Year_{Cited})$. The graph attention mechanism will be trained to parameterize a decay function, mathematically distinguishing between immediate scholarly critiques and delayed longitudinal refutations.

3.4. Formulating the Semantic hp-frac

Once the HGNN predicts the citation intent, these intents are mapped to scalar weights (e.g., Extends = 1.5, Refutes = 0.5). For any target paper p, the semantic score S(p) is the sum of its incoming weighted edges. The **Semantic hp-frac** for an author is then calculated by applying the h-index function H over the fractionally distributed semantic scores of their corpus:

Semantic hp-frac $=H([\frac{S(p_1)}{a_1}, \frac{S(p_2)}{a_2}, \frac{S(p_3)}{a_3}, \ldots, \frac{S(p_n)}{a_n}])$

(Where a_i represents the number of co-authors on paper p_i, preserving the fractional distribution of the original metric).

3.4. Formulating the True 2-Hop Semantic hp-frac

Once the HGNN predicts the citation intent and temporal decay, these values are mapped into continuous semantic edge weights (e.g., Extends = 1.5, Refutes = 0.5, scaled by the decay factor). To eliminate the semantic blindspot while preserving the mathematical rigor of multi-hop topological evaluation, we calculate the metric in two distinct phases:

Phase 1: The Paper-Level Semantic h-index (Sh_p) Existing topological models treat all citations uniformly. To resolve this, we evaluate a target paper not just by the volume of its citations, but by the semantic authority of the papers citing it. First, we calculate a base Semantic Score S© for every citing paper c by summing its own incoming weighted edges.

Then, the Semantic h-index (Sh_p) for a target paper p is defined by applying the Hirsch function to its citing network: it is the largest number h such that at least h citing papers have a Semantic Score $S(c) \geq h$. This 2-hop semantic traversal mathematically prevents artificial citation inflation; a paper is only deemed foundational if it is utilized by other methodologically sound, community-supported papers.

Phase 2: The Author-Level Semantic hp-frac The Semantic hp-frac for an author is then calculated by applying the h-index function H over the fractionally distributed Semantic h-indices of their entire published corpus:

$$Semantichp - frac = H\left(\left[\frac{Sh_{p_1}}{a_1}, \frac{Sh_{p_2}}{a_2}, \ldots, \frac{Sh_{p_n}}{a_n}\right]\right)$$

(Where a_i represents the number of co-authors on paper p_i, preserving the fractional distribution of the original metric to mitigate hyperauthorship inflation)

3.5. Distributed Systems Backend & GraphRAG UI

The computed topology and embedded node features will be stored in highly partitioned Parquet files. An Apache Spark engine will process this localized data for sub-second

dashboard metric rendering. Furthermore, the dashboard will feature a **GraphRAG** interface. When a user queries the literature, the retrieval step will strictly follow the weighted topological paths (e.g., extracting text exclusively from "Refute" edges to summarize criticisms), effectively solving the multi-hop reasoning failures of standard RAG systems.

4. Expected Contributions

1. **Algorithmic:** The formal definition and empirical validation of the Semantic hp-frac index.
2. **Architectural:** A novel integration of Section-Aware NLP, HGNNs, and Distributed Spark computing optimized for bibliometric analysis.
3. **Application:** An open-source, GraphRAG-powered dashboard capable of instant, manipulation-resistant literature evaluation.
4. **Significance of the Semantic Citation Network:** The network fundamentally transforms knowledge representation by mapping the actual qualitative dialogue of scientific research rather than just tracking static links. By classifying edge intents, differentiating between methodologies being explicitly *extended* versus datasets being *refuted*,the architecture instantly exposes citation bubbles and highly controversial paradigms that traditional graphs mask as high-impact. Furthermore, this intent-aware topology provides the structural foundation for advanced GraphRAG, enabling AI models to logically traverse specific semantic paths to generate hallucination-free, context-aware literature syntheses.
5. **Significance of the Semantic hp-frac Metric:** Building upon this infrastructure, the Semantic hp-frac metric redefines the evaluation of academic merit by eliminating the sentiment blindspot of traditional indices. By mathematically weighting citations based on their semantic intent and applying temporal decay, it creates a manipulation-resistant evaluation framework that penalizes viral but heavily criticized papers while disproportionately rewarding foundational, community-supported work. This provides the scientific community with the first dynamic, equitable metric that combines multi-hop fractional credit allocation with the true qualitative consensus of the field.

APPENDIX: Technical Execution Specifications

This section defines the strict engineering, data, and architectural requirements for implementing the Semantic Graph Pipeline and True 2-Hop Semantic hp-frac.

1. Data Engineering & The 2-Hop Expansion (The P_c^2 Layer)

To calculate the True 2-Hop metric defined in Section 3.4, the dataset must extend beyond the core papers.

- **Data Source:** Semantic Scholar Graph API.
- **Extraction Logic:** 1. Target Papers (P_w): The core dataset (e.g., SciCite training data).

2. Hop-1 Citations (P_c^1): Papers citing P_w.
3. Hop-2 Citations (P_c^2): Papers citing P_c^1.

- **Fault-Tolerant Pipeline:** The IDE must implement an API extraction script with batching (e.g., 100 IDs per request), robust rate-limit handling (HTTP 429), and disk-based checkpointing to Parquet files to prevent data loss during massive graph expansions on local or cluster hardware.
- **Node Initialization:** All paper abstracts must be vectorized using allenai/scibert_scivocab_uncased, resulting in node feature matrices of shape $[N, 768]$. Author nodes are initialized by mean-pooling the $\mathbb{R}^{768}$ abstract embeddings of the papers they have written.

2. PyTorch Geometric (PyG) Heterogeneous Schema

The IDE must construct a PyG HeteroData object with the following strict taxonomy:

- **Node Types:**
  - ['paper'].x: Tensor of shape $[N_{papers}, 768]$.
  - ['author'].x: Tensor of shape $[N_{authors}, 768]$.
- **Edge Types:**
  - ['paper', 'cites', 'paper']: Contains edge_index $[2, E_{cites}]$, edge_attr (SciBERT embeddings of citation sentences $[E_{cites}, 768]$), and delta_t (Temporal difference in publication years $[E_{cites}, 1]$).
  - ['author', 'writes', 'paper']: Contains edge_index $[2, E_{writes}]$.

3. HGNN Architecture Specifications

The neural network must be a custom dual-head Heterogeneous Graph Neural Network utilizing HeteroConv.

- **Custom Temporal Layer (TemporalGATConv):** Standard message passing layers do not support time. The IDE must write a custom layer where the attention score α between a citing paper and a cited paper is penalized by a learnable decay parameter λ: $\text{decay} = e^{-|\lambda| \Delta t}$.
- **Message Passing Routing:**
  - Author $\rightarrow$ Paper: Standard SAGEConv (passes historical domain metadata).
  - Paper $\rightarrow$ Paper: TemporalGATConv (passes semantic intent and temporal relevance).
- **Dual-Head Predictors (Outputs):**
  - *Intent Head:* A linear layer outputting 3 categorical logits (Background, Method, Result).
  - *Importance Head:* A linear layer with a Sigmoid activation outputting a continuous regression scalar (0.0 to 1.0).

4. Training Objectives & Loss Functions

- **Imbalance Handling:** The SciCite dataset is heavily skewed toward "Background" citations. The CrossEntropy loss for the Intent Head *must* implement dynamic class weighting to prevent majority-class collapse.
- **Composite Loss:** $\mathcal{L}_{Total} = \mathcal{L}_{CrossEntropy} + (0.5 \times \mathcal{L}_{BCE})$.
- **Metrics:** Track Intent Accuracy (target $>80\%$) and Importance MAE.

5. Real-Time Inference & Dashboard Architecture

The system operates in two distinct computational states:

- **The Offline Truth Engine (Batch Processing)**
    - This is where the heavy academic lifting happens. Running locally on your M4 or deployed as a batch job on a compute cluster, this pipeline runs periodically (e.g., weekly or monthly) to update the state of the scientific world.
    - The Crawl: It hits the Semantic Scholar API to fetch the missing P_c^2 boundary nodes and their abstracts.
    - The Vectorization: It passes the new abstracts through SciBERT to generate the dense embeddings.
    - The Global Inference: It loads the entire million-node graph into your trained HGNN to calculate the exact intent and temporal decay for every single edge.
    - The Math: It executes Phase 1 ($S(c)$ and Sh_p) and Phase 2 (the fractional author credit).
    - The Storage: It dumps the finalized, mathematically perfect Semantic hp-frac rankings into a columnar Parquet file.
- **Online Inductive Inference (Dashboard Upload):** When a user uploads a new PDF to the dashboard, the system *must not* recalculate the entire global graph. It parses the PDF (via GROBID), extracts the local "ego-graph" (the paper and its direct references), vectorizes the text, and runs a forward pass through the trained weights to instantly render the weighted edges in the UI.